5. Numerically solving higher order ODEs

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Recitation due Mar 6, 2024 20:30 IST

Passing to a first order system

4:28 / 4:28 2.0x

time interval zero-ten, with an initial

position of one, and velocity of zero.

All that's left is to provide the command to call ODE45, and we can run our script.

The first output from ODE45 is a vector, 'tSol', containing the times at which the numerical

solution was computed.

The second output is a matrix, 'xSol'.

The columns of 'xSol' contain the computed values of the vector components, x1 and x2.

These values in turn correspond to the position, and velocity of the mass at each time value

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Problem 2 (External resource) (1.0 points possible)

Problem 2

Next, we will practice using `ODE45` to solve the higher-order DE,

$$\ddot{x} + 2\dot{x} + x = 0, \quad x(0) = 0, \, \dot{x}(0) = 1,$$

on the interval $t \in [0, 5]$. You can verify that the analytic solution of this IVP is $x(t) = te^{-t}$. Rememl need to express this second-order DE as a system of two first-order DEs in the form $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$. To cc

1. Correctly define the variables `x0`, `tspan`, and `A`.
2. Enter an appropriate expression for the first argument of `ODE45`.
3. Define a column vector `xTrue` that contains the values of the analytic solution at the time: will need to use the $.*$ operator to perform element-wise multiplication. You can review th Operations.)

As in problem 1, if your script runs correctly you will see a plot comparing the analytic and numeric between the two solutions, and the time it took `ODE45` to compute the numerical solution. In the fir a constant time step. It begins by using a small time step but then increases the time step to reduc the error bounded.

Script

Save Reset

MATLAB Documentation (https://www.mathworks.com/help/)

```
%Numerically solve DE and time how long it takes
x0 = [0;1];
tspan = [0,5];
A = [0,1;-1,-2];
tic;
[t,x] = ode45(@(t,x) A*x,tspan,x0);
```

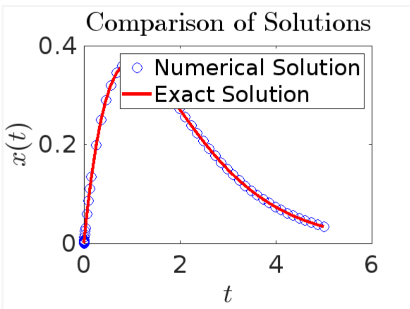
Calculator

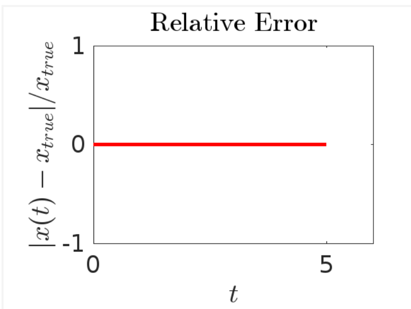
```
[t,x] = ode45(@(t,x) A*x,tspan,x0);
timeElapsed = toc;
disp(['It took ODE45 ',num2str(timeElapsed,3), ' seconds to compute the solu

%Enter analytic solution (Hint: it is in the text above.)
xTrue = x(:,1);

%Plot results
%Do not edit the code below.
figure(1)
plot(t,x(:,1),'bo','markersize',10); hold on;
plot(t,xTrue,'r','linewidth',3);
legend('Numerical Solution','Exact Solution','location','northeast');
xlabel('$t$','interpreter','latex'); ylabel('$x(t)$','interpreter','latex')
title('Comparison of Solutions','interpreter','latex')
set(gca,'fontsize',25)

figure(2)
plot(t,abs(x(:,1)-xTrue)./xTrue,'r','linewidth',3);
xlabel('$t$','interpreter','latex'); ylabel('$|x(t)-x_{true}|/x_{true}$','i
title('Relative Error','interpreter','latex')
set(gca,'fontsize',25)
```

Run Script

Output

```
It took ODE45 0.0189 seconds to compute the solution
```





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